

SPI Waveform Analysis & Protocol Timing Guide

1. Mode 0 Timing Breakdown (CPOL=0, CPHA=0)

• SCLK begins at 0 (idle low). • CS_N transitions 1 -> 0. • CRITICAL: As soon as CS_N falls, MOSI and MISO drive bit 7 (MSB) immediately before any SCLK pulse. • Leading Edge (Rising): Both Master and Slave sample bit 7. • Trailing Edge (Falling): Both Master and Slave launch bit 6. • Repeats for 8 clock cycles (16 total edges). • After 16 edges, SCLK parks at 0, CS_N stays low for T_CSH, then rises to 1.

2. Mode 1 Timing Breakdown (CPOL=0, CPHA=1)

• SCLK begins at 0 (idle low). • CS_N transitions 1 -> 0. • Leading Edge (Rising): Master and Slave launch bit 7 onto MOSI/MISO. • Trailing Edge (Falling): Master and Slave sample bit 7. • Repeats for 8 cycles. Trailing edge of cycle 8 samples bit 0 (LSB). • SCLK returns to 0, CS_N deasserts.

3. Mode 2 Timing Breakdown (CPOL=1, CPHA=0)

• SCLK begins at 1 (idle high). • CS_N transitions 1 -> 0. • Bit 7 is placed on lines at CS_N assert. • Leading Edge (Falling): Samples bit 7. • Trailing Edge (Rising): Shifts bit 6. • After 16 edges, SCLK returns to 1, CS_N returns to 1.

4. Mode 3 Timing Breakdown (CPOL=1, CPHA=1)

• SCLK begins at 1 (idle high). • CS_N transitions 1 -> 0. • Leading Edge (Falling): Launches bit 7. • Trailing Edge (Rising): Samples bit 7. • SCLK returns to 1, CS_N deasserts.